

Project Design Phase Proposed Solution Template

Date	01 October 2026
Team ID	SWTID-2026-8331
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Existing record-level access controls may be too broad because they mainly depend on user roles or permissions. The project solves this by restricting access dynamically according to a value stored in the record, so users can read or modify a record only when the configured field condition is satisfied.
2.	Idea / Solution description	Script-Controlled ACL uses a configurable script condition to evaluate a record field during an access request. The system checks the field value, applies the ACL rule, and returns an Allow or Deny decision before the requested record operation is completed.
3.	Novelty / Uniqueness	The solution combines ACL enforcement with field-value-driven scripting. Instead of relying only on static roles, administrators can define conditions that respond to the actual data in each record. This provides fine-grained, dynamic and reusable record access control.
4.	Social Impact / Customer Satisfaction	The solution helps protect sensitive records, reduce accidental data exposure and make access decisions more consistent. Clear Allow/Deny results and audit information can improve user trust while helping administrators manage access more effectively.
5.	Business Model (Revenue Model)	The solution can be offered as an enterprise security feature or integrated into an access-control platform. Potential revenue models include organizational licensing, subscription-based access-control services, enterprise support and customization for specific applications.
6.	Scalability of the Solution	The approach can be applied across different applications, databases and record types by changing the field and rule conditions. It can scale from individual applications to enterprise environments with centralized rule management, auditing and integration through APIs.